

Data Science Capstone Project EDA and Feature Engineering

Course: MSDS640 Data Science Capstone **Dataset:** UCI Adult Census Income (1994 U.S. Census)

The target is income, either over 50K or 50K and under a year, and I explore the data with my three hypotheses in mind rather than plotting for the sake of plotting. Hypothesis 1 is that education is a stronger driver of the over 50K prediction than hours worked. Hypothesis 2 is that XGBoost is the fairest of the three models toward true high earners across nativity, with a smaller US-born versus foreign-born recall gap than logistic regression and the MLP. Hypothesis 3 is that XGBoost classifies better than both the logistic baseline and the MLP on PR-AUC and F1. As in the first milestone, the data is loaded into SQLite and pulled back out before any analysis.

1. Handling missing values

Only three columns are missing anything: workclass at 5.64 percent, occupation at 5.66 percent, and native country at 1.79 percent. All three are categorical and well under the level where dropping a column would make sense. I fill each with an explicit Unknown category rather than dropping the rows, because dropping every row with a missing value would throw away roughly 2,400 people, and the missingness is not random, so dropping them could bias the income analysis. An Unknown bucket keeps the rows and lets the model treat missingness as its own signal.

I also drop the 24 exact duplicate rows while the sampling weight is still in the table, so genuinely different people are not merged by mistake. I then drop that sampling weight, which is not a fact about the person, and the text education column, which repeats the numeric one. After cleaning there are no missing values left.

2. Addressing data imbalance

The target is about 75.9 percent under 50K and 24.1 percent over 50K, roughly a 3.2 to 1 split. That drives both my choice of metrics and how I handle the imbalance.

On metrics, plain accuracy is misleading here, because a model that predicts nobody earns over 50K is right about 76 percent of the time while catching zero high earners. Precision is the share of predicted high earners who really are high earners, and recall is the share of real high earners the model actually catches. F1 combines those two, so it is more honest than accuracy when the classes are uneven. ROC-AUC can still look flattering when the positive class is small, because a large easy negative class props up the curve. PR-AUC, the area under the precision-recall curve, focuses on the positive class directly and does not get credit for the easy negatives, so it is the most reliable single number for an imbalanced problem like this. That is why my plan leads with PR-AUC and recall and treats accuracy as a secondary number.

On handling, I plan to train with class weights and with SMOTE oversampling on the training split only, and compare both against the raw imbalance. I prefer oversampling the minority class over undersampling the majority, because undersampling would throw away most of the under 50K rows and lose real information, while oversampling keeps every real row. I keep the oversampling on the training split only so the validation and test sets stay a real reflection of the true population.

3. Feature engineering and dimensionality reduction

I add three features. A native region feature groups native country into US, non-US, and Unknown, which turns a column that is about 90 percent United States into the split my fairness audit needs while keeping missing-country people separate. Two binary flags mark whether a person has any capital gain or any capital loss, because those columns are almost always zero with a few very large values, so a flag captures the wealth signal without the model over-reacting to the raw dollar amount. An hours group feature bins hours per week into ranges, which helps the education versus hours comparison in Hypothesis 1 since the effect of hours is not a straight line.

I one-hot encode the categorical columns, which gives a feature matrix of 52 columns, and I scale the numeric features so no single one dominates because of its units. In the real pipeline the scaler is fit on the training split only to avoid leaking information from the test set. A quick PCA on the numeric features spreads the variance fairly evenly across the components, so there is no small set that captures almost everything. Since there are only a handful of numeric features and Hypothesis 1 depends on knowing which original feature matters, reducing them into PCA components would hurt interpretability without buying much, so I keep the original features.

4. Showing the impact of preprocessing

Before cleaning, workclass was 5.64 percent, occupation 5.66 percent, and native country 1.79 percent missing, with 24 exact duplicate rows. After cleaning there are no missing values and none of those duplicates left.

For the class balance, I split the data once with stratification into 26,029 training rows and 6,508 test rows, then apply SMOTE to the training split only. Before SMOTE the training split is imbalanced at about 3.2 to 1, and after SMOTE the minority class is oversampled up to a 1 to 1 balance, while the test set is left untouched so it still reflects the true

population.

To show this mattered I fit a default logistic regression before preprocessing and after. Before preprocessing, on the raw imbalanced data, it reaches about 0.815 accuracy but only 0.39 recall on the over 50K class. After scaling, encoding, and oversampling, the accuracy stays about the same at 0.819 but the recall on the over 50K class jumps to 0.79, so the model now catches far more of the real high earners. This is only a sanity check with default settings, but for my lending use case that is the trade I want.

5. Summary statistics

The numeric summary shows age averaging about 38.6, education around 10 years, and hours per week about 40, while capital gain and capital loss have a median of zero and a small mean pulled up by a few large values. The frequency tables show Private is by far the most common workclass, married-civilian-spouse and never-married are the two biggest marital groups, there are roughly twice as many men as women, and the US region dominates.

For outliers I use the IQR rule rather than z-scores, because z-scores assume a roughly normal, symmetric distribution, and capital gain, capital loss, and hours worked are all heavily skewed, so a z-score would mislabel a lot of ordinary points. The IQR rule flags capital gain and hours per week the most, which is expected given how those columns pile up. Those are real values, not data errors, so I do not remove them, since removing real high earners would bias the income analysis and tree models handle skewed values without trimming.

6. Visualizations

The histograms confirm age is roughly bell shaped near 38, education clusters at high school, hours pile up at 40, and capital gain and loss are almost entirely zero. The correlation heatmap shows every off-diagonal correlation is small, so there is no strong multicollinearity, which matters for the logistic regression baseline because correlated inputs make its coefficients unstable and Hypothesis 1 leans on those coefficients.

The boxplot of education by income shows the over 50K group sits clearly higher on the education scale, though the boxes still overlap. A scatter of age against hours worked shows the over 50K points cluster in middle age and at or above full time hours, while the under 50K points are more common among the young and among people working fewer hours, with heavy overlap between the two, which says no single pair of features cleanly splits them.

7. Patterns tied to my hypotheses

For Hypothesis 1, education produces a spread of 0.741 in the over 50K rate from the lowest to the highest level, while hours produces a spread of only 0.361. Education climbs steadily and cleanly, while hours trends up but flattens at the top. This is a hint in favor of education being the stronger driver, not a final answer, because it is a one-feature-at-a-time view and education is correlated with other features. The real test is the coefficient comparison and permutation importance on the trained model.

For Hypothesis 2, I am showing the base-rate gap the models will later be audited against. US-born people earn over 50K at about 24.6 percent and foreign-born people at about 18.7 percent. If a model just learns that gap, it can end up missing real high earners among foreign-born people, which shows up as lower recall for that group, and that recall gap is what my fairness audit measures for each model. Breaking the rate out by sex shows men earn over 50K at a higher rate in both nativity groups, so nativity is not an isolated effect, which is why the plan audits sex and race alongside it.

For Hypothesis 3, the EDA shows the kind of patterns that flexible models pick up better than a straight linear model. The over 50K rate by age is a clear curve that rises through the working years and falls for the oldest ages, having any capital gain flips the rate from low to very high, which is a sharp threshold effect, and the education by marital status view shows the rate depends on the two features together rather than each one added up. Logistic regression only gets interactions if I add them by hand, while XGBoost and the MLP find them on their own, so the data is consistent with expecting XGBoost to classify better.

Overall, Hypothesis 1 gets an early hint in favor of education, Hypothesis 2 shows a real base-rate gap that holds up alongside sex, and Hypothesis 3 sees non-linear, interaction, and threshold effects that flexible models can use. I also looked for anything that cut against these three and nothing did. The main caution is the heavy class overlap on any single feature, which limits how strong each hint is but does not point any hypothesis the other way.

How I used an LLM for this assignment

I used an LLM to help create some of the visualizations in this notebook and also to help create some markdown cells in this notebook so I could correctly interpret some of the visualizations in this notebook too. I wrote and edited the analysis myself and confirmed the numbers by running the notebook top to bottom. I finally used an LLM to confirm that I covered every part of the instructions for this notebook to ensure it is fully complete.